

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 4.1: Statistics I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 4.1: Statistics I
Driving Question	DRIVING QUESTION: How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Making Sense of Chaos: Frequency Distribution Tables	Students watched a KALRO officer collect maize harvest data from 20 Nakuru County farmers and saw 20 raw values written on a chalkboard in collection order. They observed that different students got different answers when trying to find the most common value by eye, and that the raw list made it nearly impossible to see patterns or answer the county government's question about a typical harvest.	Students learned that a frequency distribution table organizes raw data by listing each unique value alongside tally marks and a final frequency count, making it possible to see at a glance how often each value occurs. They learned that tally marks are grouped in fives to make counting efficient, and that an organized frequency table reveals patterns — such as which value is most common — that are invisible in a raw unordered list.	Students explained that the KALRO officer's raw data list was difficult to interpret because it showed no organization or pattern, and that constructing a frequency distribution table transformed that chaos into a structured summary. However, they also explained that the frequency table alone still cannot give the county government the single representative number it needs — pointing forward to the need for measures of central tendency in upcoming lessons.
Lesson 2 Sorting the Story: Simple Distribution Tables & Tally Charts	Learners watched a KALRO data clerk build a frequency distribution table from a raw maize harvest list, and then constructed their own tally charts and frequency tables using daily sukuma wiki prices from Kongowea Market, Mombasa (20 market days, prices ranging from KES 10 to KES 25).	A frequency distribution table organises raw data by listing each value alongside how many times it occurs (its frequency), using tally marks as a systematic counting tool. A well-structured table makes patterns — such as the most common value and the spread of data — immediately visible in a way that a raw list does not.	Learners connected frequency tables to the Nakuru Maize Harvest Mystery by constructing a frequency table for the 20 raw harvest values, identifying what the table now reveals (e.g., the value 12 bags appeared most often) and what it still does not answer (e.g., which single number best represents all farmers — setting up the need for mean, median, and mode in upcoming lessons).
Lesson 3 Finding the Balance	Students explored the PhET 'Mean and Median' simulation and observed how the	The mean is the arithmetic average calculated by summing all values and	Students explained in two-sentence group conclusions whether the mean of 13.9 bags

Point: Mean of Ungrouped Data	mean marker (blue triangle) shifts as data points are added, removed, or dragged to extreme values. They observed the balance-beam visual mode and recorded what happens to the mean when an outlier (e.g., 200 mandazi sales) is introduced into a dataset.	dividing by the number of values ($\bar{x} = \Sigma x \div n$). It represents the 'balance point' of a dataset — the equal-redistribution value. For the 20 Nakuru maize harvest values, the mean is 13.9 bags. However, outliers (extreme values far from the rest of the data) can pull the mean away from what is truly typical, as seen in the Gikomba mandazi sales data where one day of 200 sales distorted the mean.	accurately represents the typical Nakuru farmer, connecting the balance-point concept to the anchoring phenomenon. They updated their cumulative Data Story Models with the mean marked on the number line, an equal-redistribution annotation, and a critical 'warning' note about outliers — building the conceptual foundation for comparing mean, median, and mode in Lessons 4 and 5.
Lesson 4 The Most Popular Value: Mode of Ungrouped Data	In the classroom siblings data and the mandazi market data, one value kept showing up more than all the others. In the Nakuru maize harvest data, the value 12 appeared 5 times out of 20 — more than any other value. In a bimodal data set, two values tied for most frequent. In a data set where every value is different, there is no mode at all.	The mode is the value that occurs most frequently in a data set. It can be found by building a frequency table and identifying the row with the highest frequency. A data set can be unimodal (one mode), bimodal (two modes), or have no mode. Unlike the mean, the mode is not pulled away from the typical value by outliers. For the Nakuru maize data, the mean is 13.9 bags while the mode is 12 bags.	The mode represents the most common experience in a data set, which is why it can be more useful than the mean when one typical value keeps repeating. For the Nakuru County fertilizer decision, the mode tells us what harvest level most farmers actually achieved — not an average that could be distorted by one farmer with an unusually large or small harvest. However, if the most common value is very low, using the mode might cause decision-makers to under-estimate the resources needed. This tension — and the need for a third measure — will be resolved in Lesson 5.
Lesson 5 The Middle of the Story — Median of Ungrouped Data	Learners physically lined up as data points to locate the middle value of an 11-value Nakuru harvest subset, and compared mean vs median for a Kisumu KCPE dataset with and without an outlier score.	The median is the middle value of an ordered dataset, found at position $(n+1)/2$ for odd n or by averaging the two middle values for even n ; unlike the mean, the median is not pulled away by extreme values (outliers).	For the Nakuru maize harvest data, the median (12 bags) is lower than the mean (13.9 bags) because a few farmers with very large harvests pull the mean upward — making the median a fairer representation of what a typical Nakuru smallholder farmer actually harvests, and therefore a better number for the county government to use when allocating subsidised fertiliser.
Lesson 6 Drawing the Journey — Line Graphs	We observed that monthly maize retail prices in Nairobi changed every month throughout the year, with prices lowest in the middle of the year and highest toward the end of the year, forming a visible pattern when the data points were connected.	We learned that a line graph is constructed by plotting data points on labeled axes and joining consecutive points with straight ruled lines; it shows trends, peaks, and troughs in data that changes over time, providing information that mean, mode, and median alone cannot reveal.	We explained that the KALRO officer in the Nakuru phenomenon could use a line graph of monthly maize prices to advise farmers on the best time to sell their harvest — something no single average could communicate — giving the county government richer evidence for its fertilizer allocation and food security decisions.
Lesson 7 Comparing with Bars:	Students watched a KALRO officer present a frequency distribution table of maize	Students learned that a bar graph uses rectangular bars whose heights represent	Students explained how bar graphs constructed from the Rift Valley counties'

Constructing and Interpreting Bar Graphs	harvest data from five Rift Valley counties (Nakuru, Uasin Gishu, Trans-Nzoia, Nandi, Kericho) to county government officials and observed how raw table data is difficult to interpret quickly. They also watched an instructional video demonstrating how to construct a bar graph from a frequency distribution table using the Nakuru County maize harvest data from the anchoring phenomenon.	the frequency of each category, requires labelled horizontal and vertical axes (categories and frequency respectively), a suitable scale, uniform bar widths, gaps between bars, and a descriptive title. They also learned that bar graphs enable immediate visual comparison between categories — making it possible to identify the most and least common harvest ranges at a glance — and that the tallest bar corresponds to the modal class identified in earlier lessons.	frequency distribution tables allow the KALRO officer to visually compare harvest yields across counties and identify which county's farmers harvested the least — supporting the county government's decision about where to allocate subsidized fertilizer. They also explained the connection between the mode (most frequent value, Lessons 3–6) and the tallest bar on the graph, demonstrating that different representations of the same data reveal the same underlying story in different ways.
Lesson 8 Slicing the Pie: Constructing Pie Charts from Community Data	Students observed the teacher construct a pie chart live using the Kenya National Energy Mix data (Geothermal 43%, Hydro 24%, Wind 14%, Solar 9%, Fossil Fuels & Others 10%), following each step from percentage to angle to sector drawing with compass and protractor. They replicated each step concurrently on their own circles.	Students learned that a pie chart represents each category as a sector of a circle, that sector angles are calculated using the formula $(\text{frequency} \div \text{total}) \times 360^\circ$ (or equivalently, $\text{percentage} \times 3.6^\circ$), and that all sector angles must sum to 360° . They also learned that pie charts are best suited for showing proportions across a small number of categories, and that they reveal dominance and relative share in a way that a frequency table or raw data list cannot.	Students explained how the Nakuru maize-harvest data from Lesson 1 can be converted into a pie chart that shows the proportion of farmers in each harvest group, and why this representation is useful for the KALRO officer and county government when making fertilizer allocation decisions. They also articulated at least one limitation of pie charts — that they do not reveal the spread or exact values within each category — and recorded this in their Data Story Models and on the DQB.
Lesson 9 Reading the Full Picture	Students re-watched the original anchoring phenomenon clip of the KALRO officer's chaotic chalkboard of 20 Nakuru farmers' maize harvest values, read a fictional KNBS-style 'Nakuru County Statistical Dossier' containing a pie chart of fertilizer subsidy distribution across Kenyan counties, a comparative bar graph of maize yields by Nakuru sub-county across two seasons, and a fully worked frequency distribution table of the original 20-farmer dataset — and watched a short video on the Kenya National Bureau of Statistics using data to drive national decisions.	Students learned how to read and extract meaning from multiple graphical representations simultaneously (pie charts, bar graphs, and frequency tables); how to evaluate which measure of central tendency — mean, median, or mode — is most appropriate for a given dataset and policy purpose; how to synthesize statistical evidence from multiple sources into a structured written advisory report using a Claim–Evidence–Reasoning framework; and how the entire statistical process — from raw data collection through organization, representation, and interpretation — is the foundation of real decisions made by Kenyan government bodies such as KNBS and county agricultural offices.	Students explained, through their advisory reports and final annotated models, that the chaos of the original 20-farmer harvest list (the anchoring phenomenon) can only be transformed into actionable policy advice by applying the full statistical toolkit: organizing raw data into frequency tables, calculating and comparing mean, median, and mode to identify the most representative value, and selecting appropriate graphical representations to communicate findings clearly to decision-makers. They answered the driving question — 'How can we collect, organize, and represent data about our community to help Kenyan decision-makers take effective action?' — by arguing that statistics is the systematic bridge between raw community experience and informed government action, as demonstrated through their fertilizer allocation recommendations to the fictional Nakuru

			County Director of Agriculture.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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